Activity 5.01: Clustering Sales Data Using K-Means

In this activity I have the weekly sales of 811 products over one year (52 weeks). Nobody told me which products belong together, so this is an unsupervised problem and K-means is a good fit. The idea is to let the algorithm group the products that have a similar sales pattern into 8 clusters, and then check how big each cluster is and how much it sells, so I can see which groups of products are the best sellers.

My plan:

1. Load the data
2. Look at how the columns are organized
3. Keep only the normalized columns for the clustering
4. Build and fit a K-means model with 8 clusters
5. Put the cluster labels back on the original sales data
6. Summarize each cluster: number of products, total sales and average yearly sales per product
7. Check that the numbers add up and write down what I see

Reference: So, A., So, W., & Nagy, Z. (2020). The Applied Artificial Intelligence Workshop. Packt Publishing. The dataset comes from the book's GitHub repository.

Step 1: Load the data

In [1]:
```python
import pandas as pd

# the csv is in the book's GitHub repository, so I can read it straight from the link
file_url = 'https://raw.githubusercontent.com/PacktWorkshops/The-Applied-Artificial-Intelligence-Workshop/master/Datasets/Sales_Transactions_Dataset_Weekly.csv'
df = pd.read_csv(file_url)

# first look at the table
df.head()
```

Out[1]:

	Product_Code	W0	W1	W2	W3	W4	W5	W6	W7	W8	...	Normalized 42	Normalized 43	Normalized 44	Normalized 45	Normalized 46	Normalized 47	Normalized 48	Normalized 49	Normalized 50	Normalized 51
0	P1	11	12	10	8	13	12	14	21	6	...	0.06	0.22	0.28	0.39	0.50	0.00	0.22	0.17	0.11	0.39
1	P2	7	6	3	2	7	1	6	3	3	...	0.20	0.40	0.50	0.10	0.10	0.40	0.50	0.10	0.60	0.00
2	P3	7	11	8	9	10	8	7	13	12	...	0.27	1.00	0.18	0.18	0.36	0.45	1.00	0.45	0.45	0.36
3	P4	12	8	13	5	9	6	9	13	13	...	0.41	0.47	0.06	0.12	0.24	0.35	0.71	0.35	0.29	0.35
4	P5	8	5	13	11	6	7	9	14	9	...	0.27	0.53	0.27	0.60	0.20	0.20	0.13	0.53	0.33	0.40

5 rows × 107 columns

Step 2: Understand the columns

There are a lot of columns, so before I do anything I want to know what each group is.

In [2]:
```python
# rows = products, columns = everything we know about them
df.shape
```

Out[2]:
```
(811, 107)
```

In [3]:
```python
# any missing values? If there are, k-means would complain later
int(df.isnull().sum().sum())
```

Out[3]:
```
0
```

In [4]:
```python
# I print the names without the repeated W1..W51 and Normalized 1..51 to keep it short
cols = df.columns.tolist()
print(cols[:3], '...', cols[51:56], '...', cols[-2:])
```

```
['Product_Code', 'W0', 'W1'] ... ['W50', 'W51', 'MIN', 'MAX', 'Normalized 0'] ... ['Normalized 50', 'Normalized 51']
```

So the 107 columns are:

- `Product_Code` : the name of the product (P1, P2, ...)
- `W0` to `W51` : how many units were sold in each of the 52 weeks (these are the raw sales)
- `MIN` and `MAX` : the lowest and highest weekly sales of that product
- `Normalized 0` to `Normalized 51` : the same weekly sales, but scaled for each product between 0 and 1 using its own `MIN` and `MAX`

There are no missing values, so I don't need to clean anything.

Let me quickly check that last point about the normalized columns, because it matters for the next step.

In [5]:
```python
normalized_cols = [f'Normalized {i}' for i in range(52)]

# if each product is scaled with its own min and max, every row must have a 0 and a 1 somewhere
print('every product has a 0 :', (df[normalized_cols].min(axis=1) == 0).all())
print('every product has a 1 :', (df[normalized_cols].max(axis=1) == 1).all())
```

```
every product has a 0 : True
every product has a 1 : True
```

Step 3: Keep only the normalized columns

For the clustering I'm going to use the normalized columns and not the raw ones. If I used the raw sales, a product that sells 70 units a week would always look very different from one that sells 5 a week, and K-means would mostly be splitting products by size. With the normalized data, what counts is the *shape* of the sales over the year (when it goes up and down), not how big the numbers are. The size of the sales is something I look at afterwards, when I compare the clusters.

The book does this by dropping the first 55 columns by position. I'd rather pick the columns by name, because then I can see exactly what I'm keeping.

In [6]:
```python
# df2 has just the 52 normalized columns, one row per product
df2 = df[normalized_cols]
df2.shape
```

Out[6]:
```
(811, 52)
```

Step 4: Build and fit the K-means model

The activity asks for 8 clusters, so `n_clusters=8`.

In [7]:
```python
from sklearn.cluster import KMeans

# 8 clusters, same random_state as the book, 10 random starts
k_means_model = KMeans(n_clusters=8, random_state=8, n_init=10)

# fit only learns from the normalized data
k_means_model.fit(df2)
```

```
C:\Users\sabre\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=4.
  warnings.warn(
```

Out[7]:
```
▾ KMeans ⓘ ?
▸ Parameters
▸ Fitted attributes
▸ 8 features
```

In [8]:
```python
# labels_ has the cluster number (0 to 7) of every product, in the same order as the rows
labels = k_means_model.labels_
labels[:20]
```

Out[8]:
```
array([7, 7, 3, 7, 7, 0, 7, 3, 3, 3, 7, 3, 4, 0, 4, 4, 4, 4, 4, 3],
      dtype=int32)
```

In [9]:
```python
# how many products ended up in each cluster?
pd.Series(labels).value_counts().sort_index()
```

Out[9]:
```
0    127
1     54
2    115
3     98
4    125
5     89
6     84
7    119
Name: count, dtype: int64
```

The cluster numbers (0 to 7) are only names. Cluster 3 is not "better" than cluster 2, and if I ran this on a different version of the library the same group of products could get a different number.

Step 5: Add the labels to the sales data

Now I want to go back to the raw sales so I can see how much each cluster actually sells. I take the product code and the 52 weekly columns and add the label as a new column. I make a copy so that `df` itself stays as it was.

In [10]:
```python
weekly_cols = [f'W{i}' for i in range(52)]

# product code + weekly sales only, the MIN/MAX and normalized columns are not needed anymore
sales = df[['Product_Code'] + weekly_cols].copy()

# same row order as df2, so the labels line up with the right products
sales['label'] = labels
sales.head()
```

Out[10]:

	Product_Code	W0	W1	W2	W3	W4	W5	W6	W7	W8	...	W43	W44	W45	W46	W47	W48	W49	W50	W51	label
0	P1	11	12	10	8	13	12	14	21	6	...	7	8	10	12	3	7	6	5	10	7
1	P2	7	6	3	2	7	1	6	3	3	...	4	5	1	1	4	5	1	6	0	7
2	P3	7	11	8	9	10	8	7	13	12	...	14	5	5	7	8	14	8	8	7	3
3	P4	12	8	13	5	9	6	9	13	13	...	10	3	4	6	8	14	8	7	8	7
4	P5	8	5	13	11	6	7	9	14	9	...	11	7	12	6	6	5	11	8	9	7

5 rows × 54 columns

Step 6: Summarize each cluster

For every cluster I want three things: how many products it has, how many units it sold in total over the 52 weeks, and the average yearly sales per product (total divided by number of products). I first add up the 52 weeks for every product, and then group by cluster.

In [11]:
```python
# total units sold over the whole year, for each product
sales['total_sales'] = sales[weekly_cols].sum(axis=1)

# one row per cluster: number of products and total sales
df_final = sales.groupby('label').agg(
    count_product=('Product_Code', 'count'),
    total_sales=('total_sales', 'sum'),
)

# average yearly sales per product in that cluster
df_final['yearly_average_sales'] = df_final['total_sales'] / df_final['count_product']

# best selling clusters first
df_final = df_final.sort_values(by='yearly_average_sales', ascending=False)
df_final
```

Out[11]:

label	count_product	total_sales	yearly_average_sales
4	125	180199	1441.592000
7	119	71217	598.462185
3	98	50656	516.897959
0	127	57363	451.677165
5	89	11493	129.134831
1	54	1704	31.555556
6	84	1895	22.559524
2	115	760	6.608696

Step 7: Check the numbers

Before I trust the table I want to be sure nothing got lost when grouping. Two simple checks: all 811 products should be in some cluster, and the cluster totals should add up to the total sales of the whole dataset.

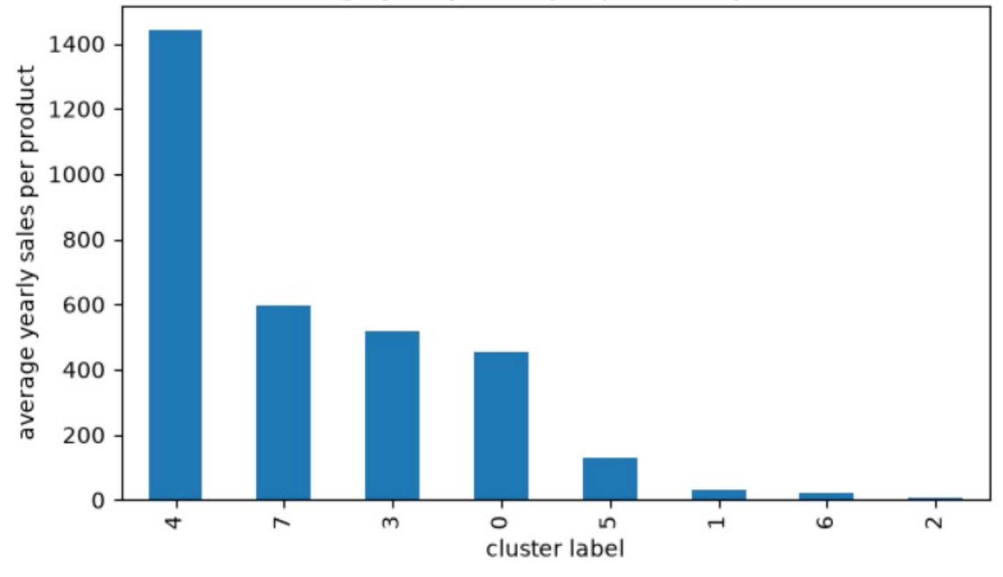
In [12]:
```python
print('products in clusters :', df_final['count_product'].sum(), '| products in the data:', len(df))
print('sales in clusters    :', df_final['total_sales'].sum(), '| sales in the data   :', df[weekly_cols].sum().sum())
```

```
products in clusters : 811 | products in the data: 811
sales in clusters    : 375287 | sales in the data   : 375287
```

Both match, so the table is complete. Now a small chart of the last column, to see the differences between clusters more easily than with numbers.

In [13]:
```python
import matplotlib.pyplot as plt

# bar chart of the average yearly sales per product in each cluster
df_final['yearly_average_sales'].plot(kind='bar', figsize=(7, 4))
plt.xlabel('cluster label')
plt.ylabel('average yearly sales per product')
plt.title('Average yearly sales per product by cluster')
plt.show()
```



Conclusion

In my run the table shows a big difference between the clusters. The best cluster (label 4) has 125 products that sold about 180,000 units in total, which is around 1,440 units per product in the year. At the other end, the weakest cluster (label 2) has 115 products with only about 760 units in total, or roughly 7 units per product. So the best group sells more than 200 times as much per product as the weakest one.

To put it in perspective, I can calculate the share of the total sales for each cluster:

In [14]:
```python
# what percent of all products and of all sales does each cluster have?
share = pd.DataFrame({
    'pct_of_products': (df_final['count_product'] / df_final['count_product'].sum() * 100).round(1),
    'pct_of_sales': (df_final['total_sales'] / df_final['total_sales'].sum() * 100).round(1),
})
share
```

Out[14]:

label	pct_of_products	pct_of_sales
4	15.4	48.0
7	14.7	19.0
3	12.1	13.5
0	15.7	15.3
5	11.0	3.1
1	6.7	0.5
6	10.4	0.5
2	14.2	0.2

The top cluster is about 15% of the products but brings in about 48% of all the units sold. The three clusters at the bottom (labels 1, 6 and 2) are roughly a third of the catalog (253 products, about 31%) and together they add up to only about 1% of the sales (4,359 units out of 375,287). For a store this is useful: it shows which products carry the business and which ones barely sell, and those would be the first to look at when deciding what to promote or what to stop stocking.